MoM1

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In **Section 1** we have derived the Electric Field Integral Equation (EFIE) which is the starting point for all our subsequent work. The derived EFIE equation can serve for general purposes each time no equivalence volume or surface theorems are applied, i.e., where no magnetic currents are designed for the model under analysis [2]. We recall this equation here:

$$[j\omega\mu(\mathcal{L}\mathbf{J})(\mathbf{r})]_{tan} = [\mathbf{E}^i(\mathbf{r})]_{tan}, \tag{1}$$

where $\mathcal{L}$ is the following integro-differential operator:

$$\mathcal{L}\mathbf{X}(\mathbf{r}) = \int_V G(\mathbf{r},\mathbf{r}')\left[1 + \frac{1}{k^2}\nabla'\nabla'\cdot\right]\mathbf{X}(\mathbf{r}')d\mathbf{r}', \tag{2}$$

and $G(\mathbf{r},\mathbf{r}')$ is the Green function of the scalar Helmholtz equation

$$\nabla^2 G(\mathbf{r},\mathbf{r}') + k^2 G(\mathbf{r},\mathbf{r}') = -\delta(\mathbf{r},\mathbf{r}'). \tag{3}$$

In most applications, it is customary to regard the current density $\mathbf{J}$ as unknown, while the incident field $\mathbf{E}^i$ as given. The Method of Moments (MoM) is a numerical method which enables us to turn the EFIE equation into a finite dimensional linear system of several unknowns from which one can easily reconstruct an approximation of the current density $\mathbf{J}$. This is accomplished by using a set of *basis* functions, chosen carefully to mimic the local behaviour of the expected current distribution, and by using a set of *weighting* functions, which make possible to derive a finite linear system from the approximation made in the continuum and to reduce the residual among approximate boundary conditions and exact boundary conditions on the whole domain of interest. We now present the general formulation of the method.

1.1 The Method of Moments

Consider any given linear operator $\mathbf{S}$ that acts in some function space $\mathcal{H}$ and ranges in some other function space $\mathcal{K}$. Assume $D(\mathbf{S}) \subseteq \mathcal{H}$ to be the domain of the operator $\mathbf{S}$. Let g be in the range $\mathrm{Ran}(\mathbf{S})$ and let $f \in D(\mathbf{S})$ be such that

$$\mathbf{S}(f) = g. \tag{4}$$

The objective is to approximate the function f once the forcing function g is known. In electrodynamical problems dealing with scattering, typically the operator $\mathbf{S}$ is an integro-differential operator, while the forcing therm is a driving function, e.g., the incident electric field in the EFIE. To achieve this, we expand f in terms of basis functions capable of accurately representing and resemble the unknown function. Among them, two classes of basis functions are generally used in MoM: *subdomain* functions and *entire domain* functions. In the first class, we find compact domain functions and they are used in such a way that the computational domain is meshed and covered with them; hence, each one provides a local approximation of the physical quantity of interest. The second class, the functions of the entire domain, is rarely used in practice, but their role is analogous to that of trigonometric polynomials in the approximation of functions by truncated Fourier series.

Let $\chi_1, \chi_2, \ldots, \chi_N$ be a set of basis functions. We require that the unknown f can be expanded in terms of these functions as

$$f(\mathbf{r}) \approx \sum_{n=1}^{N} a_n \chi_n(\mathbf{r}), \tag{5}$$

and by the linearity of $\mathbf{S}$, we have the following:

$$\sum_{n=1}^{N} a_n \mathbf{S}(\chi_n)(\mathbf{r}) \approx g(\mathbf{r}). \tag{6}$$

The above equation is a linear equation in N unknowns $a_1, a_2, \ldots a_N$. The aim of the Method of Moments is to generate a linear system where all a_j's can be determined at once by solving the linear system. To do this, we define a set of N weighting functions $w_1, w_2, \ldots, w_N$ lying in the domain of the operator $\mathbf{S}$ and make N scalar products to obtain the following linear system:

$$\begin{cases} a_1\langle w_1, \mathbf{S}(\chi_1)\rangle + a_2\langle w_1, \mathbf{S}(\chi_2)\rangle + \cdots + a_N\langle w_1, \mathbf{S}(\chi_N)\rangle = \langle w_1, g\rangle \\ \\ a_1\langle w_2, \mathbf{S}(\chi_1)\rangle + a_2\langle w_2, \mathbf{S}(\chi_2)\rangle + \cdots + a_N\langle w_2, \mathbf{S}(\chi_N)\rangle = \langle w_2, g\rangle \\ \\ \qquad\qquad \cdots \qquad\qquad \cdots \qquad\qquad \cdots \\ \\ a_1\langle w_N, \mathbf{S}(\chi_1)\rangle + a_2\langle w_N, \mathbf{S}(\chi_2)\rangle + \cdots + a_N\langle w_N, \mathbf{S}(\chi_N)\rangle = \langle w_N, g\rangle \end{cases} \tag{7}$$

This procedure is general in nature. A common choice for weighting functions is that of Dirac distributions $w_k(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r_k})$, where $\mathbf{r}_k$'s with $k = 1, \ldots, N$, are arbitrary points of the computational domain of the unknown f. This choice is often referred to as *point-matching* (or *collocation*) and it is

generally employed in applications. This is analogous in choosing arbitrary N points $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N$ in the computational domain, and then the following linear system:

$$\begin{cases} a_1\mathbf{S}(\chi_1)(\mathbf{r}_1) + a_2\mathbf{S}(\chi_2)(\mathbf{r}_1) + \cdots + a_N\mathbf{S}(\chi_N)(\mathbf{r}_1) = g(\mathbf{r_1}) \\ \\ a_1\mathbf{S}(\chi_1)(\mathbf{r}_2) + a_2\mathbf{S}(\chi_2)(\mathbf{r}_2) + \cdots + a_N\mathbf{S}(\chi_N)(\mathbf{r}_2) = g(\mathbf{r_2}) \\ \\ \qquad\qquad \ldots \qquad\qquad \ldots \qquad\qquad \ldots \\ \\ a_1\mathbf{S}(\chi_1)(\mathbf{r}_N) + a_2\mathbf{S}(\chi_2)(\mathbf{r}_N) + \cdots + a_N\mathbf{S}(\chi_N)(\mathbf{r}_N) = g(\mathbf{r_N}) \end{cases} \tag{8}$$

with compact form

$$\mathbf{Za} = \mathbf{g}, \qquad Z_{mn} = \mathbf{S}(\chi_n)(\mathbf{r}_m). \tag{9}$$

This is the procedure and choice we have made in subsequent work for the solution of the EFIE of general wires. Moreover, a suitable set of basis functions is then used, that is, the basis of *pulse functions*. These functions are precisely the characteristic functions of subdomains of the Euclidean spaces:

$$\chi_A(\mathbf{r}) = \begin{cases} 1 & \text{whenever } \mathbf{r} \in A \\ 0 & \text{whenever } \mathbf{r} \in A^c \end{cases} \tag{10}$$

Therefore, it is clear that mathematically a better approximation of the unknown function is obtained by a denser subdivision of the domain of the fields and densities under investigation, although in practice this is not generally the case due to other factors, which could degrade the final solution as we will shortly see. For the sake of completeness, it is also necessary to point out that the choice of the pulse functions is not the only one possible: in practice there is an extensive use of piece-wise linear functions or piece-wise sinusoidal functions, which furnish under some circumstances a better rate of convergence for the unknown. For a better understanding of this problem and others (e.g. other sets of weighting functions and basis functions, Garlekin's method and convergence issues), we refer to [3] and [1].

1.2 Application to several unknowns

Referring to the EFIE equation, the electrodynamical system under analysis could commonly be composed of several elements such as linear, circular, and other conductors (or even dielectrics if the appropriate integro-differential operator is defined) with their specifics in terms of conduction parameters. It is interesting and will be useful later to explicitly include this differentiation in the MoM formulation. For example, consider a set $\Omega_1, \Omega_2, \ldots, \Omega_M$ of conductors or dielectrics, and let $\mathbf{r}^{(1)}, \mathbf{r}^{(2)}, \ldots, \mathbf{r}^{(M)}$ denote the corresponding variables that

label the computational spaces of each one of them. The total current density $\mathbf{J}$ that permeates the space is then given by the vector summation of the current densities $\mathbf{J}^{(1)}, \mathbf{J}^{(2)}, \ldots, \mathbf{J}^{(M)}$ lying on the objects $\Omega_1, \Omega_2, \ldots, \Omega_M$. By the linearity of the operator $\mathcal{L}$ we are led to the equation

$$\sum_{k=1}^{M} [j\omega\mu(\mathcal{L}\mathbf{J}^{(k)})(\mathbf{r}^{(k)})]_{tan} = \sum_{k=1}^{M} [\mathbf{E}^i(\mathbf{r}^{(k)})]_{tan}. \tag{11}$$

Having motivated the multi-object formulation in the electromagnetic setting, we now return to the abstract operator notation introduced above in the hypothesis that the total unknown function f can be written as a summation of several other unknown functions $f^{(1)}, f^{(2)}, \ldots, f^{(M)}$. As before, the corresponding variables $\mathbf{r}^{(1)}, \mathbf{r}^{(2)}, \ldots, \mathbf{r}^{(M)}$ will label the computational domain of them. The equation we want to discretize is

$$\sum_{k=1}^{M} \mathbf{S}(f^{(k)}) = g. \tag{12}$$

We now expand each $f^{(k)}$, $k = 1, \ldots, M$, with a family of basis functions

$$f^{(k)} \approx a_1^{(k)}\chi_1^{(k)} + a_2^{(k)}\chi_2^{(k)} + \cdots + a_{N_k}^{(k)}\chi_{N_k}^{(k)}. \tag{13}$$

We notice that, in general, the basis $\{\chi_n^{(k)}\}_{n=1}^{N_k}$ is different from $\{\chi_n^{(h)}\}_{n=1}^{N_h}$ for $k \neq h$: they could differ for the number of elements N_k and N_h and also for the type of functions involved. We will turn the above equation into a linear system using the point-matching procedure. For each of the variables $\mathbf{r}^{(1)}, \mathbf{r}^{(2)}, \ldots, \mathbf{r}^{(M)}$ we chose the right amount of points which is equal to the number of basis functions implied in the approximation of each of the $f^{(1)}, f^{(2)}, \ldots, f^{(M)}$. The system developed in this way will be a square system and can be written as follows:

$$\begin{cases} \displaystyle\sum_{k=1}^{M}\sum_{n=1}^{N_k} a_n^{(k)}\mathbf{S}(\chi_n^{(k)})(\mathbf{r}_1^{(1)}) = g(\mathbf{r}_1^{(1)}) \\ \qquad\qquad \cdots \qquad\qquad\qquad \cdots \\ \displaystyle\sum_{k=1}^{M}\sum_{n=1}^{N_k} a_n^{(k)}\mathbf{S}(\chi_n^{(k)})(\mathbf{r}_1^{(1)}) = g(\mathbf{r}_{N_M}^{(M)}) \end{cases} \tag{14}$$

We can organize the unknowns and the forcing term as follows:

$$\mathbf{a}^{(k)} = \begin{pmatrix} a_1^{(k)} \\ a_2^{(k)} \\ \cdots \\ a_{N_k}^{(k)} \end{pmatrix}, \qquad \mathbf{g}^{(k)} = \begin{pmatrix} g(\mathbf{r}_1^{(k)}) \\ g(\mathbf{r}_2^{(k)}) \\ \cdots \\ g(\mathbf{r}_{N_k}^{(k)}) \end{pmatrix}. \tag{15}$$

and the system can be written in the following compact form as a block system:

$$\begin{pmatrix} \mathbf{Z}_{1,1} & \mathbf{Z}_{1,2} & \dots & \mathbf{Z}_{1,M} \\ \mathbf{Z}_{2,1} & \mathbf{Z}_{2,2} & \dots & \mathbf{Z}_{2,M} \\ \dots & \dots & \ddots & \dots \\ \mathbf{Z}_{M,1} & \mathbf{Z}_{M,2} & \dots & \mathbf{Z}_{M,M} \end{pmatrix} \begin{pmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \dots \\ \mathbf{a}^{(M)} \end{pmatrix} = \begin{pmatrix} \mathbf{g}^{(1)} \\ \mathbf{g}^{(2)} \\ \dots \\ \mathbf{g}^{(M)} \end{pmatrix} \tag{16}$$

where the matrix elements of the blocks are given by

$$(\mathbf{Z}_{h,k})_{l,m} = \mathbf{S}(\chi_m^{(k)})(\mathbf{r}_l^{(h)}). \tag{17}$$

The block $\mathbf{Z}_{h,k}$ has dimensions $N_h \times N_k$, $h, k = 1, \dots, M$, therefore, in general only the terms of the type $\mathbf{Z}_{h,h}$ will be square. By setting $N_{tot} = \sum_{k=1}^{M} N_k$, the total matrix, which we will simply denote by $\mathbf{Z}$, has dimension $N_{tot} \times N_{tot}$ and is square. The blocks $\mathbf{Z}_{h,k}$ will be called *self-terms* if $h = k$ or *coupling-terms* for $h \neq k$. The self-term $\mathbf{Z}_{h,h}$ represents the effect of the basis functions of $f^{(h)}$ on the equations tested on the computational domain of $f^{(h)}$, e.g., the action of each part of Ω_h on each part of itself. The coupling-term $\mathbf{Z}_{h,k}$, for $h \neq k$, represents the effect of the basis functions of $f^{(k)}$ on the equations tested on the computational domain of $f^{(h)}$.

We notice that the full matrix $\mathbf{Z}$ can be written as a contribution of two terms

$$\begin{pmatrix} \mathbf{Z}_{1,1} & 0 & \dots & 0 \\ 0 & \mathbf{Z}_{2,2} & \dots & 0 \\ \dots & \dots & \ddots & \dots \\ 0 & 0 & \dots & \mathbf{Z}_{M,M} \end{pmatrix} + \begin{pmatrix} 0 & \mathbf{Z}_{1,2} & \dots & \mathbf{Z}_{1,M} \\ \mathbf{Z}_{2,1} & 0 & \dots & \mathbf{Z}_{2,M} \\ \dots & \dots & \ddots & \dots \\ \mathbf{Z}_{M,1} & \mathbf{Z}_{M,2} & \dots & 0 \end{pmatrix} \tag{18}$$

where the first is the block matrix of self-terms and the second is the block matrix of coupling-terms. This distinction will be extremely useful in the following chapters, where self- and coupling-terms may require different numerical treatments.

The formulation developed in this section provides the general mathematical framework that will be used throughout this work. Starting from continuous operator equation, the Method of Moments leads, through the choice of basis functions and testing points, to a finite-dimensional linear system. The same construction naturally extends to systems involving several interacting domains whose mutual interactions are represented by the off-diagonal blocks of the resulting matrix. In the following sections, this general framework will be specialized to the EFIE for a perfectly conducting thin wire. In this setting, abstract unknown coefficients acquire the physical meaning of current coefficients and the matrix $\mathbf{Z}$ becomes the impedance matrix of the discretized electromagnetic problem.

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In this section, we specialize the theory of the Method of Moments introduced in the previous section to a *perfectly electric conductor* (PEC) *thin wire*.

Consider a perfectly straight conductor wire in a medium of electrical permittivity ϵ and magnetic permeability μ, oriented along the $\hat{\mathbf{z}}$ axis and with radius a. Fix the operative wavelength and the length L of the wire. The thin-wire approximation requires

$$a \ll L, \qquad a \ll \lambda. \tag{19}$$

An incident external electric field $\mathbf{E}^i$ excites the wire and produces a surface current density $\mathbf{J}_s$ on it. Under these assumptions, if the wire is very thin, the current density $\mathbf{J}_s$ is not a function of the azimuthal angle and is predominately axial. Therefore, we have

$$I(z)\hat{\mathbf{z}} = \int_0^{2\pi} \mathbf{J}_s(z)\, a\, d\phi, \tag{20}$$

from which we obtain

$$\mathbf{J_s} = \frac{I(z)}{2\pi a}\hat{\mathbf{z}}, \tag{21}$$

where $I(z)$ denotes the total axial current flowing through the cross section of the wire at z.

2.1 Pocklington's equation

The volume integral in Eq.(2) reduces to a surface integral:

$$\mathcal{L}\mathbf{J}_s(\mathbf{r}) = \left(\int_{-L/2}^{L/2}\int_0^{2\pi} \frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}\left[1+\frac{1}{k^2}\frac{\partial^2}{\partial z'^2}\right]\frac{I(z')}{2\pi}d\phi' dz'\right)\hat{\mathbf{z}}, \tag{22}$$

where the function

$$G(\mathbf{r},\mathbf{r}') = \frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}, \tag{23}$$

is the Green function of the scalar Helmholtz Eq.(3). Notice that the quantity $|\mathbf{r}-\mathbf{r}'|$ can be expressed in cylindrical coordinates as

$$|\mathbf{r}-\mathbf{r}'| = \sqrt{(z-z')^2 + |\rho-\rho'|^2}. \tag{24}$$

Taking $\rho' = a$, one has

$$|\rho-\rho'| = \rho^2 + a^2 - 2\rho a\cos(\phi'-\phi), \tag{25}$$

and the above distance can be expressed as

$$|\mathbf{r}-\mathbf{r}'| = \sqrt{(z-z')^2 + \rho^2 + a^2 - 2\rho a\cos(\phi'-\phi)}. \tag{26}$$

If we assume a to be very small, we can approximate this distance as follows:

$$|\mathbf{r} - \mathbf{r}'| \approx R = \sqrt{(z - z')^2 + \rho^2}. \tag{27}$$

We remark that the above approximation does not imply that the physical radius is set to zero. The radial coordinate of the observation point is retained, and, when the boundary condition is enforced by locating on the surface of the wire, it is set $\rho = a$. Consequently, the reduced distance entering the kernel becomes

$$R = \sqrt{(z - z')^2 + a^2}, \tag{28}$$

so that the finite wire radius is still accounted for the resulting thin-wire formulation. This makes it possible to perform the integration in the ϕ' variable in Eq.(22). Integration by parts then yields

$$\mathcal{L}\mathbf{J}_s(\mathbf{r}) = \left(\int_{-L/2}^{L/2} I(z') \left[1 + \frac{1}{k^2} \frac{\partial^2}{\partial z'^2} \right] \frac{e^{-jkR}}{4\pi R} dz' \right) \hat{\mathbf{z}}, \tag{29}$$

leading to the following EFIE equation, where E_z^i is the tangential component of the incident field on the PEC conductor:

$$j\omega\mu \int_{-L/2}^{L/2} I(z') \left[1 + \frac{1}{k^2} \frac{\partial^2}{\partial z'^2} \right] \frac{e^{-jkR}}{4\pi R} dz' = E_z^i(z). \tag{30}$$

This last equation is generally called *Pocklington's integral equation* and will be the starting point of our subsequent analysis. [3, 2, 1].

2.1.1 Hallen's equation

For the sake of completeness, we remark here that Eq.(30) is not the only equation capable of describing the behaviour of the current on a thin-wire. in **Section 1** we have shown that the EFIE equation can also be written with the differential operator in front of the integral as

$$j\omega\mu \left[1 + \frac{1}{k^2} \nabla\nabla \right] \int_V G(\mathbf{r}, \mathbf{r}') \mathbf{J}(\mathbf{r}') d\mathbf{r}' = \mathbf{E}^i(\mathbf{r}). \tag{31}$$

By applying the same approximations as above, the EFIE for the thin-wire model reduces to

$$j\omega\mu \left[1 + \frac{1}{k^2} \frac{\partial^2}{\partial z^2} \right] \int_{-L/2}^{L/2} I(z') \frac{e^{-jkR}}{4\pi R} dz' = E_z^i(z), \tag{32}$$

which is generally called *Hallén's integral equation* [3]. Although Pocklington's equation exhibits less favorable properties than Hallén's equation, owing to the action of the differential operator on the Green's function (we refer to [3] for convergence rate comparison), we will use Pocklington's equation throughout this work because its numerical discretization by the MoM techniques is

exceptionally easy: the only preliminary step is the computation of the second derivative of the Green function. In contrast, in the formulation considered here, the application of the MoM to the Hallen's equation requires the Green's function associated with the differential operator

$$\mathcal{D} = j\omega\left[1 + \frac{1}{k^2}\frac{\partial^2}{\partial z^2}\right], \tag{33}$$

and then to derive a closed form of the vector potential. The derivation of this alternative formulation is outside the scope of the present work

2.2 Modelling the excitations

To solve Pocklington's equation, the excitation term on its right-hand side, that is, the incident field E_z^i, must be specified. In the transmitting case considered here, the incident field is therefore replaced by an impressed field representing the antenna feeding mechanism. Since detailed electromagnetic modeling of a physical feeding structure would unnecessarily complicate the thin-wire problem, an idealized feed model is introduced. The two most common and widely used feed models are the *delta-gap* and the *magnetic frill*.

2.2.1 Delta-gap

Suppose that the dipole is in transmission mode with the feed in its center at $z = 0$ and its terminals at distance Δ_g. Using the delta-gap, it is assumed that the excitation voltage at the input terminals is constant V, and that the impressed electric field is a uniform field within the gap and zero elsewhere. This field is given by

$$E_z^i(z) = \begin{cases} \frac{V}{\Delta_g}, & \text{for } -\Delta_g/2 \leq z \leq \Delta_g/2 \\ \\ 0 & \text{elsewhere} \end{cases} \tag{34}$$

2.2.2 Magnetic frill

At its center, the feed gap is replaced with a circumferential magnetic current density that exists over an annular aperture of inner radius a, the radius of the wire, and outer radius $b > a$. Over the annular aperture, the magnetic current density is given by

$$\mathbf{M}(\rho) = -\frac{V}{\rho \ln(\frac{b}{a})}\hat{\phi}. \tag{35}$$

from which one can find the excitation E_z^i as

$$E_z^i(z) = \frac{V}{2\ln(\frac{b}{a})}\left(\frac{e^{-jkR_1}}{R_1} - \frac{e^{-jkR_2}}{R_2}\right), \tag{36}$$

where

$$R_1 = \sqrt{z^2 + a^2}, \tag{37}$$

and

$$R_2 = \sqrt{z^2 + b^2}. \tag{38}$$

For the complete derivation of the excitation field of the magnetic frill, we refer to [1]. We note that the magnetic current $\mathbf{M}$ that occurs here is the Maxwell equivalent of the electric field inside a coaxial line that feeds a monopole antenna over an infinite ground plane. We refer to [3, 2, 1] for details.

Although the magnetic frill model generally provides a more realistic representation of the feeding region and may improve the computation of the input impedance, the delta-gap model provides an adequate description of the radiation characteristics. The numerical comparison between the delta-gap and the magnetic frill approximations (and particularly their influence on the computed feed current and input impedance) will be investigated in the subsequent numerical analysis.

2.3 MoM discretization of Pocklington's equation

We will now apply the Method of Moments to the Eq.(30). Before doing this, it is useful to compute directly the explicit expression of the kernel K entering the integral equation, that is

$$K(z, z') = \left[1 + \frac{1}{k^2}\frac{\partial^2}{\partial z'^2}\right]\frac{e^{-jkR}}{4\pi R}. \tag{39}$$

By performing the two derivates, its expression is given by

$$K(z, z') = \frac{e^{-jkR}}{4\pi R}\left[1 - \frac{j}{kR} - \frac{1}{k^2R^2} - (z - z')^2\left(\frac{1}{R^2} - \frac{3j}{kR^3} - \frac{3}{k^2R^4}\right)\right]. \tag{40}$$

We refer to [2, 1] for its derivation. This formula can be used to compute the matrix elements of the MoM matrix $\mathbf{Z}$ as will be shown below.

Let I be the filamentary current on the dipole, which is assumed to be of length L. We divide the dipole into N segments of lengths $\Delta z = \frac{L}{N}$. These segments are identified with subsets $\ell_1, \ell_2, \ldots \ell_N$ of the z axis as

$$\ell_n = \left(z : -\frac{L}{2} + (n-1)\Delta z < z < -\frac{L}{2} + n\Delta z\right). \tag{41}$$

We approximate the current I using the basis of pulse functions:

$$I(z) \approx \sum_{n=1}^{N} a_n \chi_n(z), \tag{42}$$

where

$$\chi_n(z) = \begin{cases} 1 & \text{whenever } z \in \ell_n \\ 0 & \text{whenever } z \notin \ell_n^c \end{cases} \tag{43}$$

The continuous equation Eq.(30) is then discretized as

$$j\omega\mu \sum_{n=1}^{N} a_n \int_{-L/2}^{L/2} \chi_n(z') K(z, z') dz' \approx E_z^i(z), \tag{44}$$

where the integrals are simply given as

$$\int_{-L/2}^{L/2} \chi_n(z') K(z, z') dz' = \int_{\ell_n} K(z, z') dz'. \tag{45}$$

taking into account the definition of χ_n in Eq.(43). Then we apply the collocation procedure: we consider the middle point of each segment $\ell_1, \ell_2, \ldots \ell_N$ and call them $z_1, z_2, \ldots, z_N$. these are given by

$$z_n = -\frac{L}{2} + \left(n - \frac{1}{2}\right)\Delta z, \qquad n = 1, \ldots, N. \tag{46}$$

The MoM system takes the form

$$\begin{cases} j\omega\mu \sum_{n=1}^{N} a_n \int_{\ell_n} K(z_1, z') dz' = E_z^i(z_1) \\ \\ j\omega\mu \sum_{n=1}^{N} a_n \int_{\ell_n} K(z_2, z') dz' = E_z^i(z_2) \\ \\ \qquad \ldots \qquad \ldots \qquad \ldots \\ \\ j\omega\mu \sum_{n=1}^{N} a_n \int_{\ell_n} K(z_N, z') dz' = E_z^i(z_N) \end{cases} \tag{47}$$

which in compact form is simply written as

$$\mathbf{Za} = \mathbf{V}, \tag{48}$$

where

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \cdots \\ a_N \end{pmatrix}, \qquad \mathbf{V} = \begin{pmatrix} E_z^i(z_1) \\ E_z^i(z_2) \\ \cdots \\ E_z^i(z_N) \end{pmatrix}, \tag{49}$$

and the matrix elements are given as follows:

$$Z_{m,n} = j\omega\mu \int_{\ell_n} K(z_m, z')dz'. \tag{50}$$

In applications, matrix elements $Z_{m,n}$ can be numerically evaluated using quadrature or other approximation methods using the explicit expression in Eq.(40).

The forcing term $\mathbf{V}$ depends on the type of excitation modeled. Assume that a delta-gap source is located in the center of the dipole with a gap width of Δz and a subdivision of the wire into an odd number N of segments. Using Eq.(34), the forcing term in Eq.(48) takes the form of

$$\mathbf{V}_\Delta = \begin{pmatrix} 0 \\ \cdots \\ 0 \\ \frac{V}{\Delta z} \\ 0 \\ \cdots \\ 0 \end{pmatrix} \tag{51}$$

where the only non-vanishing component is the $\frac{N+1}{2}$-component which is equal to $\frac{V}{\Delta z}$. In contrast, if a magnetic frill is implemented, then the collocation procedure gives for the n-component of the excitation field the following:

$$(\mathbf{V}_{MF})_n = \frac{V}{2\ln(\frac{b}{a})}\left(\frac{e^{-jkR_{1,n}}}{R_{1,n}} - \frac{e^{-jkR_{2,n}}}{R_{2,n}}\right), \tag{52}$$

where

$$R_{1,n} = \sqrt{z_n^2 + a^2}, \qquad R_{2,n} = \sqrt{z_n^2 + b^2}. \tag{53}$$

The excitations $\mathbf{V}_{\boldsymbol{\Delta}}$ and $\mathbf{V}_{MF}$ are extremely different in nature: the former is localized in a very thin region, while the latter is distributed throughout the length of the dipole. This suggests that the solution of the linear system in Eq.(48) may depend on the choice of the excitation mode used. In fact, while the discretization is refined, the term in Eq.(51) blows up and produces an increasing peak for the solution of the current localized in the feeding position, as will be observed in the numerical analysis of the present work. This is not generally a problem for the computation of radiating quantities as *gain*, *front-back ratio*, or *side-lobe levels* (which may remain comparatively stable) but it can negatively affect the computation of the *input impedance*. These and other problems will be addressed in the following sections, where we will compare the two feed models for the thin wire and Yagi-Uda configurations of wires.

2.4 Electromagnetic field reconstruction

In this section, we furnish all the expressions for the electromagnetic field generated by the current distribution of a thin wire. These formulas have been obtained in the continuum and will be used for the derivation of the *radiation pattern* as well as for all radiative quantities of interest, once the MoM solution is obtained for the current $I(z)$.

We start from the constitutive equations derived in **Section 1**. The vector potential generated by the filamentary current $I(z)$ of a thin wire is given by

$$\mathbf{A}(\mathbf{r}) = \left(\mu \int_{-L/2}^{L/2} I(z') \frac{e^{-jkR}}{4\pi R} dz'\right) \hat{\mathbf{z}} \tag{54}$$

where

$$R = \sqrt{(z-z')^2 + \rho^2} = \sqrt{(z-z')^2 + x^2 + y^2}, \tag{55}$$

and the magnetic and electric fields radiated in the time harmonic regime are given, respectively, by

$$\mathbf{H}(\mathbf{r}) = \frac{1}{\mu} \nabla \times \mathbf{A}(\mathbf{r}), \tag{56}$$

$$\mathbf{E}(\mathbf{r}) = -j\omega \mathbf{A}(\mathbf{r}) - \frac{j}{\omega \epsilon \mu} \nabla (\nabla \cdot \mathbf{A})(\mathbf{r}). \tag{57}$$

2.4.1 Electromagnetic kernels

By performing the long computations in the constitutive electromagnetic equations above, i.e., computing the curl, divengence and the gradient of the vector potential $\mathbf{A}$, one is led to consider the following integral kernels:

$$K_x^H(\mathbf{r}, z') = -\frac{e^{-jkR}}{4\pi R} \left(\frac{jky}{R} + \frac{y}{R^2}\right), \tag{58}$$

$$K_y^H(\mathbf{r}, z') = \frac{e^{-jkR}}{4\pi R} \left(\frac{jkx}{R} + \frac{x}{R^2}\right), \tag{59}$$

$$K_x^E(\mathbf{r}, z') = \frac{e^{-jkR}}{4\pi R} \left[\frac{j}{\omega \epsilon}(z-z') \left(\frac{k^2 x}{R^2} - \frac{3jkx}{R^3} - \frac{3x}{R^4}\right)\right], \tag{60}$$

$$K_y^E(\mathbf{r}, z') = \frac{e^{-jkR}}{4\pi R} \left[\frac{j}{\omega \epsilon}(z-z') \left(\frac{k^2 y}{R^2} - \frac{3jky}{R^3} - \frac{3y}{R^4}\right)\right], \tag{61}$$

$$K_z^E(\mathbf{r}, z') = \frac{e^{-jkR}}{4\pi R} \left\{-j\omega\mu + \frac{j}{\omega\mu} \left[\frac{jk}{R} + \frac{1}{R^2} + (z-z')^2 \left(\frac{k^2}{R^2} - \frac{3jk}{R^3} - \frac{3}{R^4}\right)\right]\right\}. \tag{62}$$

Setting $\mathbf{K}^E$ and $\mathbf{K}^H$ as follows:

$$\mathbf{K}^H(\mathbf{r}, z') = K_x^H(\mathbf{r}, z')\, \hat{\mathbf{x}} + K_y^H(\mathbf{r}, z')\, \hat{\mathbf{y}} \tag{63}$$

$$\mathbf{K}^E(\mathbf{r}, z') = K_x^E(\mathbf{r}, z')\,\hat{\mathbf{x}} + K_y^E(\mathbf{r}, z')\,\hat{\mathbf{y}} + K_z^E(\mathbf{r}, z')\,\hat{\mathbf{z}}, \tag{64}$$

the electromagnetic field is simply given as

$$\mathbf{H}(\mathbf{r}) = \int_{-L/2}^{L/2} I(z')\mathbf{K}^H(\mathbf{r}, z')dz', \tag{65}$$

$$\mathbf{E}(\mathbf{r}) = \int_{-L/2}^{L/2} I(z')\mathbf{K}^E(\mathbf{r}, z')dz'. \tag{66}$$

These last two expressions can be evaluated by numerical integration in each point $\mathbf{r} = (x, y, z)$ of interest using the integral kernels in Eqs.(58, 59, 60, 61, 62). In particular, suppose $\mathbf{r}_0$ to be any observation point, and suppose that we have solved the MoM system in Eq.(47). The current $I(z)$ is then approximated in terms of pulse basis functions as

$$I(z) \approx \sum_{n=1}^{N} a_n \chi_n(z). \tag{67}$$

Taking into consideration the definition in Eq.(43), the electromagnetic field can then be computed numerically as

$$\mathbf{H}(\mathbf{r}_0) \approx \sum_{n=1}^{N} a_n \int_{\ell_n} \mathbf{K}^H(\mathbf{r}_0, z')dz', \tag{68}$$

$$\mathbf{E}(\mathbf{r}_0) \approx \sum_{n=1}^{N} a_n \int_{\ell_n} \mathbf{K}^E(\mathbf{r}_0, z')dz'. \tag{69}$$

2.4.2 Cartesian to spherical components

In radiation problems, it is often necessary to express the electromagnetic field in spherical coordinates to compute the *radiation pattern* or other antenna parameters such as *front-back ratio*, *side-lobe level*, *band-width*, *directivity* and others. Moreover, in the *far field region*, some of the spherical components of the electromagnetic field radiated by a dipole drop faster than others: in the far region, for the magnetic field, the spherical component which predominates is the azimuthal one, while for the electric field is the co-latitude component. This can also be seen from the rapid investigation of the electromagnetic kernels given in the previous section by letting $R \to \infty$.

Therefore, let $\mathbf{F}$ be a vector field with Cartesian components F_x, F_y and F_z. Consider the change of coordinates $\Phi : (r, \varphi, \theta) \to (x, y, z)$ given by

$$\begin{cases} x = r\cos(\varphi)\sin(\theta) \\ y = r\sin(\varphi)\sin(\theta) \\ z = r\cos(\theta) \end{cases} \qquad r \in (0, +\infty),\ \varphi \in [0, 2\pi),\ \theta \in [0, \pi). \tag{70}$$

The local change of basis induced by this transformation furnishes the rule to obtain the spherical components of the field

$$\begin{pmatrix} F_r \\ F_\varphi \\ F_\theta \end{pmatrix} = \begin{pmatrix} \cos(\varphi)\sin(\theta) & \sin(\varphi)\sin(\theta) & \cos(\theta) \\ -\sin(\varphi) & \cos(\varphi) & 0 \\ \cos(\varphi)\cos(\theta) & \sin(\varphi)\cos(\theta) & -\sin(\theta) \end{pmatrix} \begin{pmatrix} F_x \\ F_y \\ F_z \end{pmatrix}. \tag{71}$$

This rule can be applied to the fields in Eqs.(68,69) to obtain their spherical counterparts.

The content of this section allows for the electromagnetic field analysis of the dipole, i.e., the analysis of the electromagnetic field in the near, intermediate, and far regions, and to compute radiative parameters with the aid of numerical integration techniques. For a detailed analysis of the *near region*, *intermediate region* and *far region*, we refer to [2].

3 CH5

This section is devoted to the numerical analysis and validation of the *Method of Moments* formulation (Eq.(47)) developed in the preceding chapter, considering both delta-gap and magnetic frill excitation modes. We will mainly answer the following questions:

- Does the MoM solver provide physically reliable current distributions?
- How do the results depend on the refinement of the discretization, the radius of the wire, and the excitation mode?
- Are the electromagnetic quantities coherent with the theoretical predictions and the conservation of energy?
- Within which parameter ranges can we trust the solver?

3.1 Experimental protocol

We define here below the reference configuration for the dipole that will be used as a common condition for the experiments. Any change which may occur in the parameters will be immediately indicated.

Parameters	**Values**
Wavelength λ	0.60 m
Dipole length L	0.30 m
Radius a	0.3 mm or 3 mm
Conductor	PEC
Nominal Voltage	1 V
Feed models	Delta-gap, magnetic frill
Magnetic frill ratio $\frac{b}{a}$	$\sqrt{1.1}$
Discretization	Uniform, N odd
Numerical method	Pulse basis, point matching
External reference	NEC2, HFSS

Table 1: Reference configuration

The work frequency of the dipole is 500 MHz and the radius a above will be chosen to be 0.3 mm or 3 mm, identifying two distinct configurations. We will subdivide our numerical simulations as follows:

1. **Current**

   Comparison of the simulated current distributions for the two excitation models. Stability of the current: local oscillations. Quantitative stabilization of the current: validation with NEC2 and HFSS.

2. **Input Impedance**

   Comparison of the simulated input impedance for the two excitation models. Sensitivity of the impedance. Quantitative stabilization of the input impedance: validation with NEC2 and HFSS.

3. **Electromagnetic field reconstruction and radiation**.

   Asymptotic behaviour of the electromagnetic field. Radiation pattern, directivity, and gain. Stability of radiative quantities and power consistency: validation with NEC2 and HFSS.

4. **Numerical reliability and validity domain**

3.2 Current

The analysis of this section regards the behaviour of the current distribution in both cases of delta-gap and magnetic frill excitations.

3.2.1 Delta-gap

For the delta-gap and radius $a = 0.3$ mm, the numerical analysis developed here shows that by increasing the number of subdivisions for the wire, the current approaches a limiting profile. For $N \approx 201$ the profile does not change much as N increases. These observations suggest an apparent stabilization of MoM solution which occurs over the investigated discretization range, for this wire configuration.

Things are quite different for the case of radius $a = 3$ mm. In this case, the current develops an oscillatory behaviour in the vicinity of feeding point for increasing N. Moreover, this behaviour is extremely pronounced from $N \approx 201$ on, and assumes a very pronunced peak at the feed point for $N = 401$. Despite the increasingly pronounced oscillations near the feed, the current distribution away from the excitation region remains comparatively regular

This shows that the numerical behaviour of the computed current distribution depends significantly on the radius of the wire. In particular, it is not always possible to obtain a global stabilization of the current distribution by simply increasing the number of segments, but this possibility is strongly related to the magnitude of the radius.

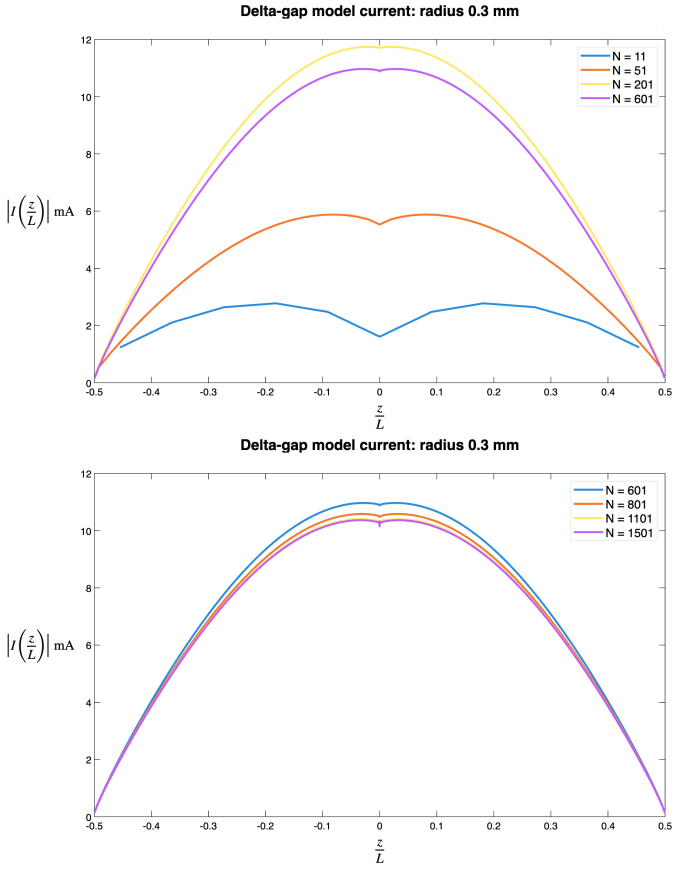


Figure 1: Radius $a = 0.3$ mm

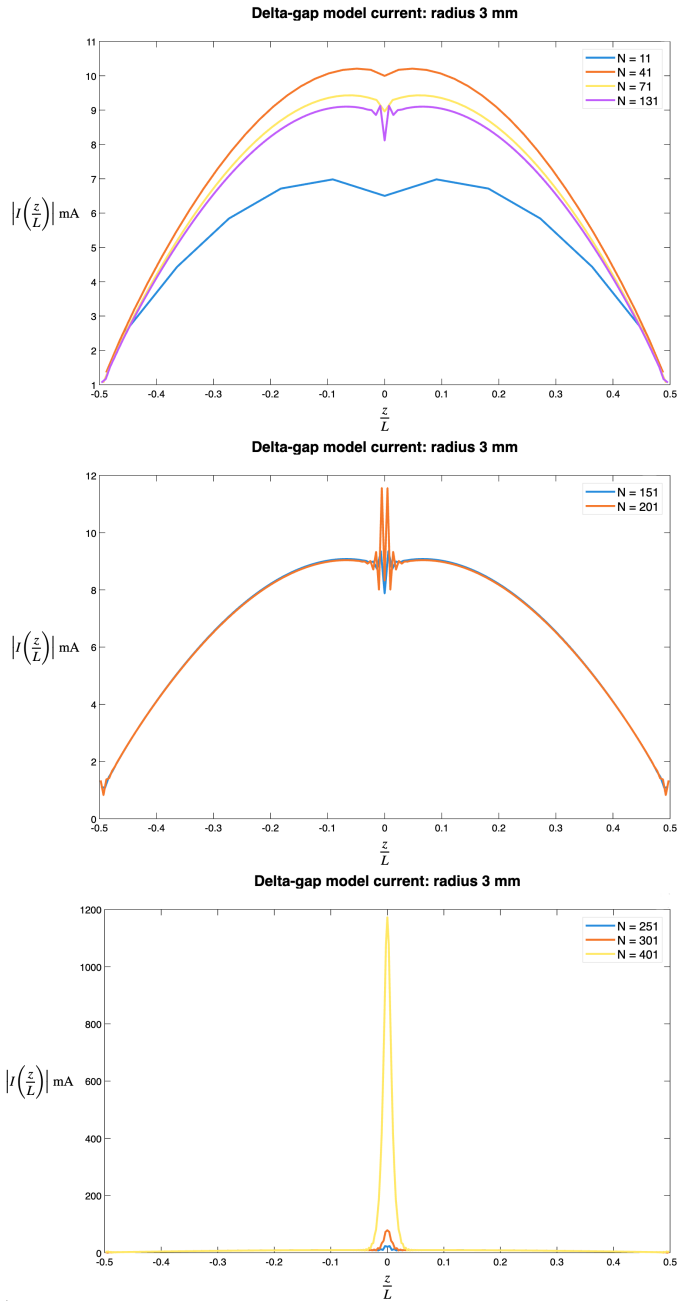


Figure 2: Radius $a = 3$ mm

This preliminary analysis suggests that the growth of anomalous feed current onset depends on the radius of the wire. It is interesting to plot the $|I(0)|$-feed

current versus the radius of the wire. The adopted radius spans an interval from $a_{min} = 10^{-4}$ to $a_{max} = 5 \cdot 10^{-3}$. As will be seen in the picture, the anomalous growth is also dependent on the discretization adopted.

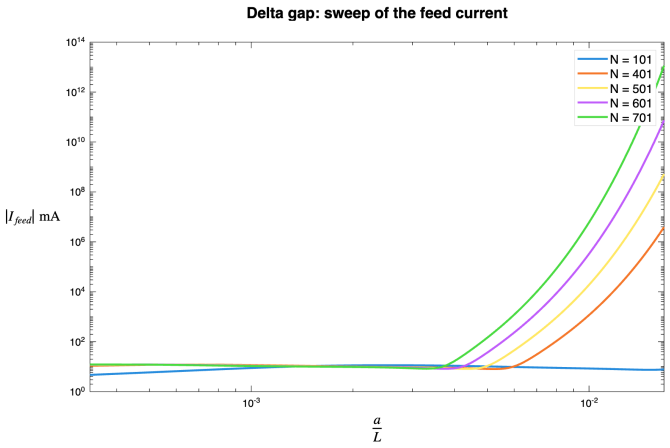


Figure 3: N = 101, 401, 501, 601, 701

Although the dimensionless ratio $\frac{a}{L}$ provides a natural parameter for characterizing this behaviour, the transition curves do not collapse onto a single universal profile. This suggests that the observed numerical instability cannot be characterized solely by the segment-length-to-radius ratio. We show here below the dependence of the I_{feed}-current on the discretization parameter N and the radius a, and plot the level curves of the surface representing it:

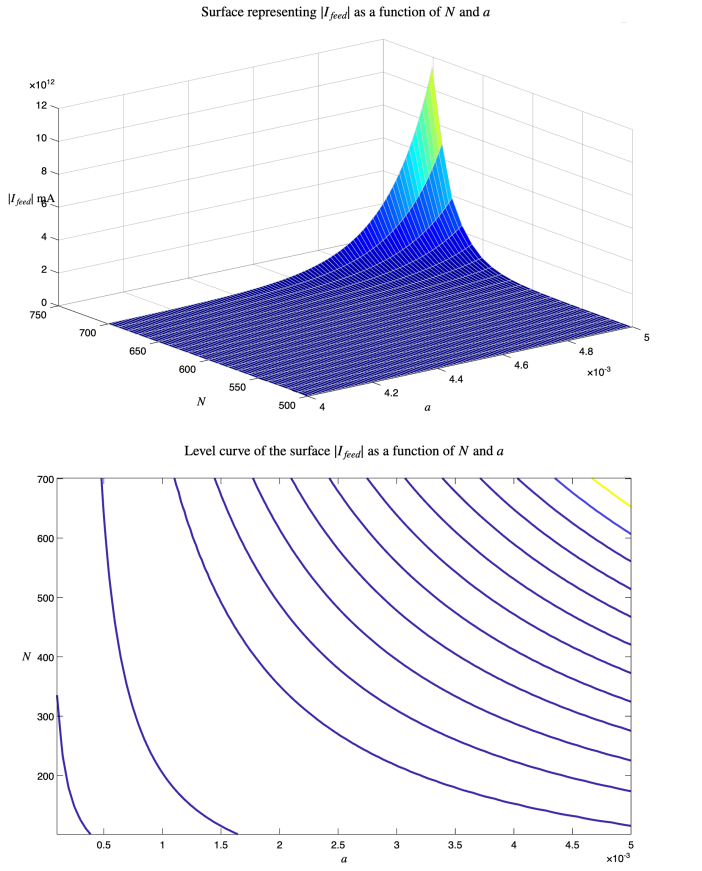


Figure 4: Dependence of $|I_{feed}|$ on the discretization and radius

These graphs show that for a given fixed value of the feed current $I_0 = |I_{feed}|$, the radius of the wire that accompanies the occurrence of this feed current decreases with the discretization number N, hence a functional relation of the type

$$a_{I_0}(N) = C(I_0)N^{-\alpha(I_0)}, \tag{72}$$

occurs, but it strongly depends on the threshold I_0.

This picture suggests that the fit law could be derived for any given I_0. For example, taking $I_0 = 14$ mA, the standard polynomial fit gives us the value of $\alpha(14 \text{ mA}) \simeq 0.8577$ and $C(14 \text{ mA}) \simeq 0.3234$ with a very good agreement between the fit slope and the data:

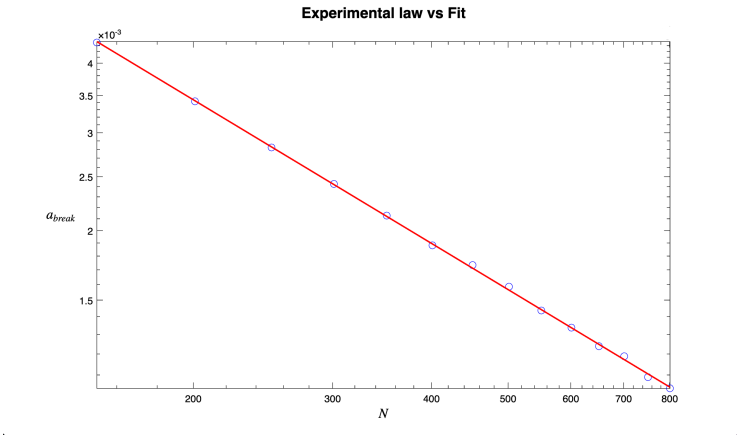


Figure 5: Fit: case of $I_0 = 14$ mA

The agreement between the numerical data and the fitted curve supports the existence of an approximate power-law dependence of $a_{I_0}(N)$ on the number of segments N, at least within the discretization range investigated.

In particular, the estimated exponent $\alpha(14\text{ mA}) \simeq 0.8577$ indicates that the radius decreases more slowly than the length of the segment $\Delta z = L/N$. Indeed, the fitted law implies

$$\frac{a_{I_0}(N)}{\Delta z} = \frac{C(14\text{ mA})}{L} N^{1-\alpha(14\text{ mA})} \propto N^{0.1423}. \tag{73}$$

Consequently, the critical ratio between the wire radius and the segment length is not constant but exhibits a slow increase as the discretization is refined, at least in the discretization region explored. This results show that increasing the number of segments does not necessarily improve the numerical behaviour of the computed feed current for a fixed wire radius.

Nevertheless, the scaling law obtained here is empirical and should not be interpreted as a universal stability criterion. Its validity remains restricted to the investigated discretization range, the adopted delta-gap formulation, and the operational breakdown threshold $I_0 = 14\,\text{mA}$.

3.2.2 Magnetic frill

In this section, we investigate the magnetic frill excitation and compare its numerical analysis with that of the delta-gap excitation mode.

For $a = 0.3$ mm, the rate of stabilization of the magnetic frill seems to be slower compared to the delta-gap: there is a transition from $N \approx 201$ to $N \approx 601$ with a change of $|I(0)|$ from ≈ 35 mA to ≈ 12 mA, against the change of ≈ 1 mA, in the same rage of discretization and for the same quantity, in the delta-gap excitation mode. From $N \approx 601$ on, it appears that the magnetic frill

finally reaches a region of the discretization parameter N, where the current exhibits the almost same profile.

In the case of a radius of $a = 3$ mm, the global convergence seems to be faster than in the case of $a = 0.3$ mm, as occurs for the delta-gap excitation model. This model of excitation does not develop the oscillatory behaviour at the feed which characterizes the delta-gap, even for high values of the discretization parameter N. Despite the good behaviour at the feed location, things are not as well behaved at the dipole ends. From $n \approx 301$ on, very huge oscillations occur at its edges, showing that the current is numerically unstable there, for large N parameters.

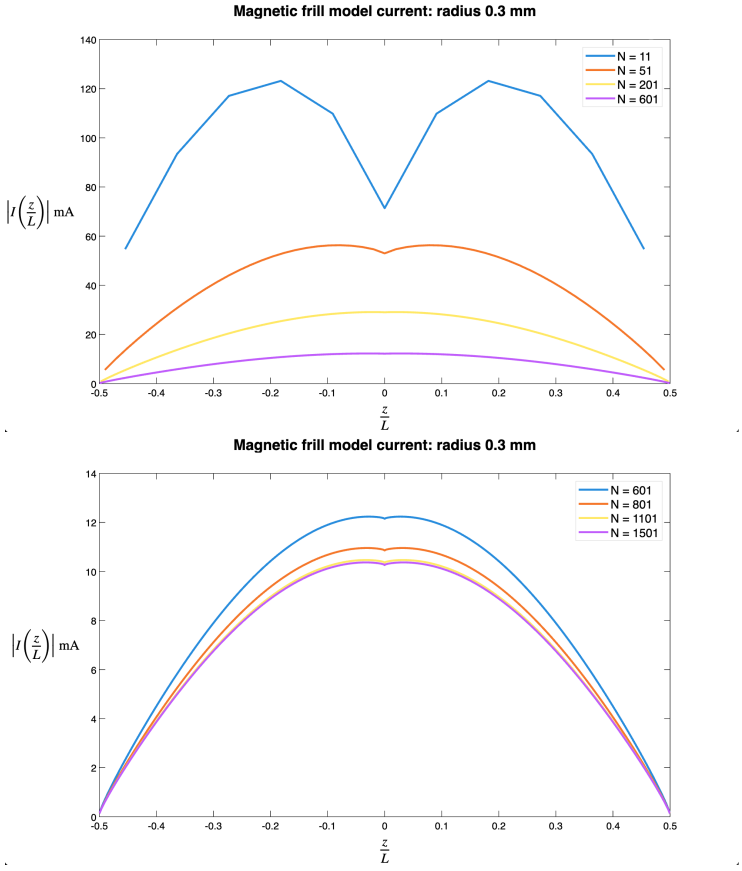


Figure 6: Radius $a = 0.3$ mm

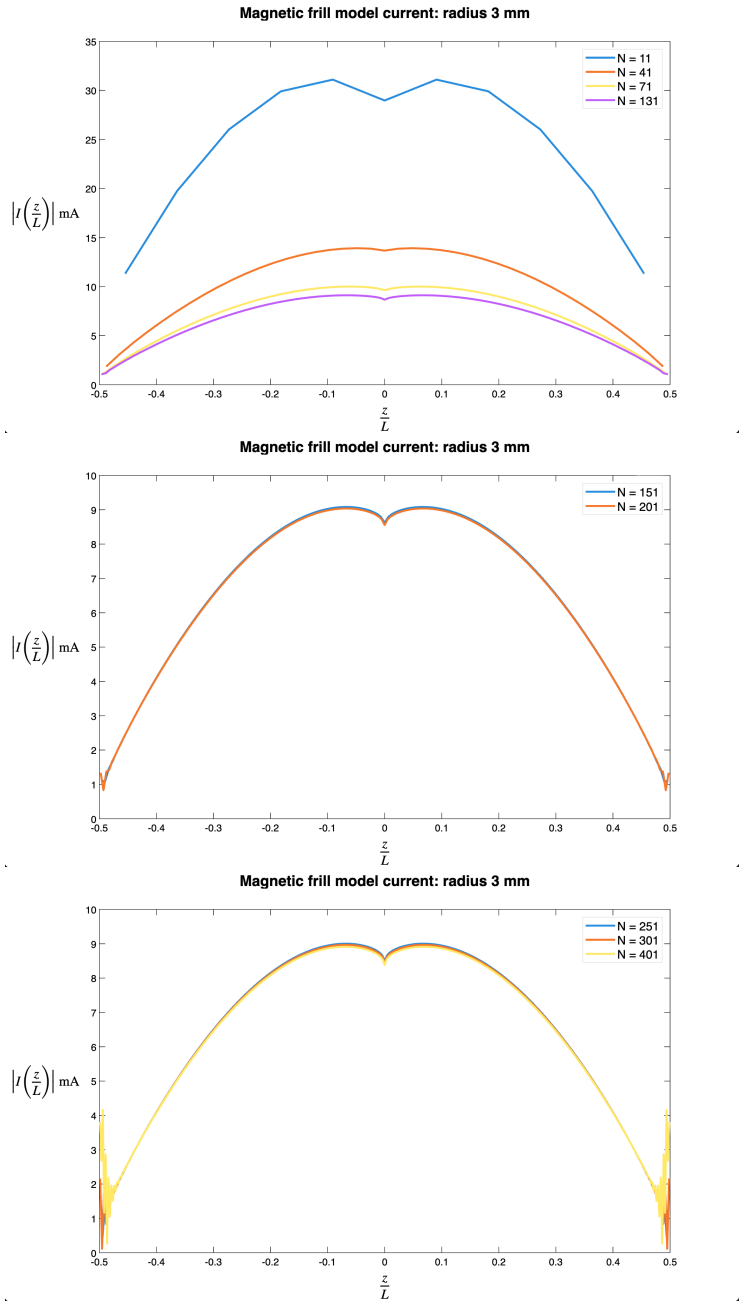


Figure 7: Radius $a = 3$ mm

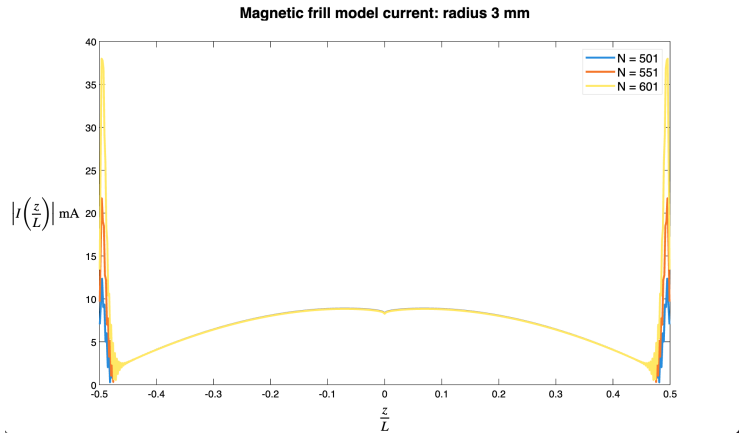


Figure 8: N = 501, 551, 601 for radius $a = 3$ mm

As for the delta-gap source, numerical instabilities appear to be strictly related to the radius of the wire. Therefore, it is interesting to perform a sweep of the endpoint current $|I(-L/2+\Delta z/2)|$ against the ratio $\frac{a}{L}$. The investigated radii are $a \in [10^{-4}, 5\cdot 10^{-3}]$ and the outer radius of the magnetic frill excitation is chosen to be $b = \sqrt{1.1}\, a$ for each a.

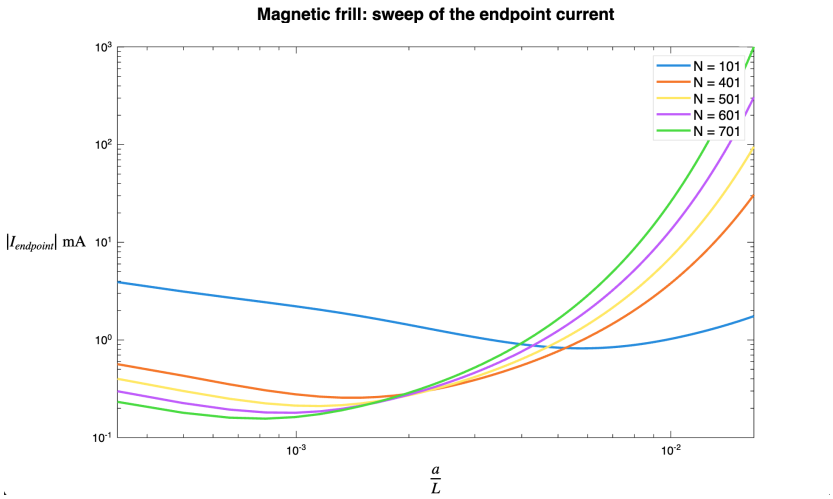


Figure 9: N = 101, 401, 501, 601, 701

We notice that the numerical instability depends on the discretization adopted. We can also plot the dependence of the $I_{endpoint}$-current on the discretization parameter N and the radius a, and plot its level curves:

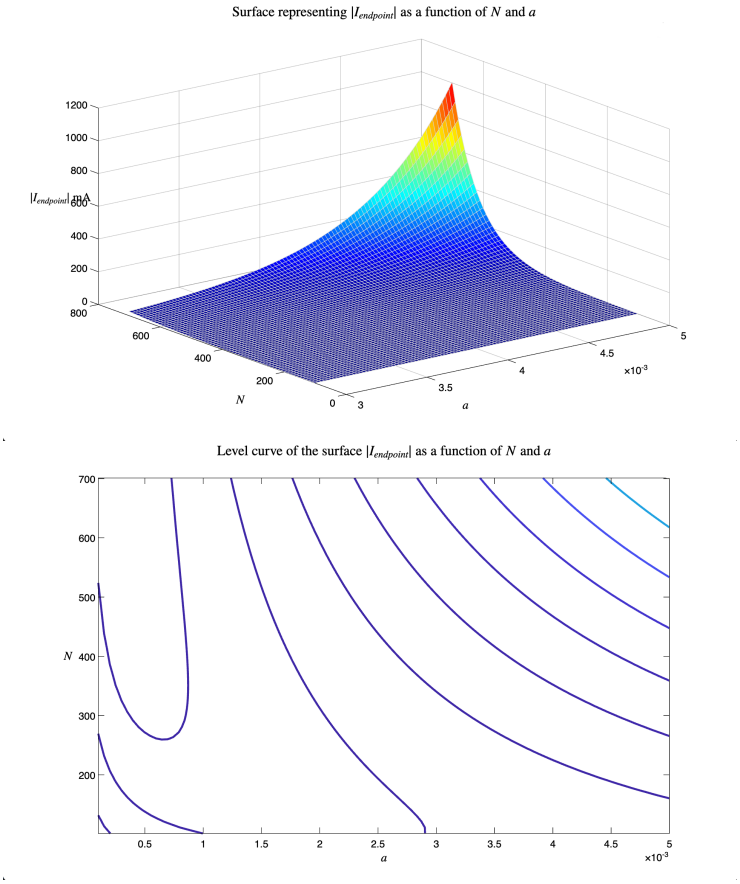


Figure 10: Dependence of $|I_{endpoint}|$ on the discretization and radius

As occurs for the delta-gap feed instability, for fixed $I_1 = |I_{endpoint}|$, in the magnetic frill there is a functional correspondence between the radius $a_{I_1}(N)$ and the discretization parameter N. In particular, this dependence is not always monotonic, but it depends on the threshold I_1. For high values of I_1, or for high value of the N parameter, the dependence is monotonically decreasing and the suggested rule is

$$a_{I_1}(N) = D(I_1)N^{-\beta(I_1)}. \tag{74}$$

For example, taking $I_1 = 1$ mA, a polynomial fit gives $\beta(10 \text{ mA}) \simeq 0.6670$ and $D(1 \text{ mA}) = 0.0975$ with a good agreement between the theoretical slope and the data:

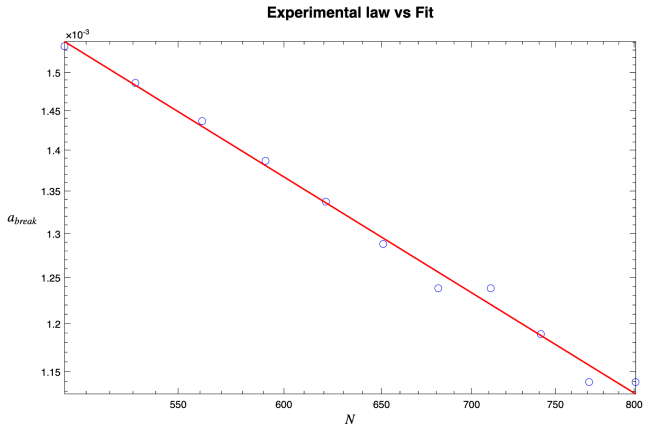


Figure 11: Fit: case $I_1 = 1$ mA

The estimated exponent indicates that the radius of decreases significantly more slowly than the length of the segment $\Delta z = L/N$. This law implies

$$\frac{a_{I_1}(N)}{\Delta z} = \frac{D(I_1)}{L} N^{1-\beta(I_1)} \propto N^{0.3330}. \tag{75}$$

Consequently, the critical ratio between the radius of the wire and the length of the segment increases with the refinement of the mesh, consistent with the dependence observed in the formulation of the delta-gap.

It should be considered that the scaling law obtained here is empirical and should not be interpreted as a universal stability criterion. Its validity remains restricted to the investigated discretization range, the adopted delta-gap formulation, and the operational breakdown threshold $I_0 = 1\,\mathrm{mA}$.

3.3 Current validation with NEC2 and HFSS

A comparison between the two excitation models reveals an important difference in their numerical behaviour. Although the delta-gap formulation exhibits an anomalous growth of the current in the vicinity of the feeding point, the magnetic frill formulation shows a comparatively regular current distribution near the feed, with numerical instabilities developing predominantly near the wire endpoints.

This section is devoted to the validation of the two models by using NEC2 for the delta-gap excitation and HFSS for the magnetic frill. The scope is to determine the range of accuracy of our solver with respect to the external program used and to establish an empirical law which relates this range to the radius of the wire.

3.3.1 Delta-gap validation: $a = 0.3$ mm

The following picture represents profiles of the current distribution given by NEC2 for some values of the discretization parameters N. The source is located at the center segment:

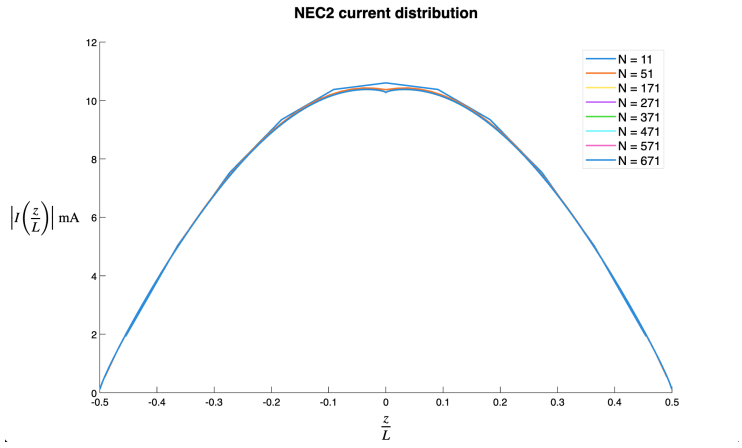


Figure 12: NEC2 current profile

We notice the fast stabilization of the current provided by NEC2. We will consider as a reference current distribution the NEC2 profile with $N^{\text{ref}} = 371$ segments. This current will be called I^{ref} in the subsequent analysis.

In the following pictures, we can see that the solution provided by out MoM solver stabilizes around the profile of I^{ref}. Moreover, other graphs are given for the ℓ^2-norm error of our discretization and the I^{ref} current. In particular, for any N adopted, we have first considered a spline interpolation of the I^{ref} current, and then we have analyzed the following two relative errors:

$$\epsilon_{\ell^2}(N) = \sqrt{\frac{\sum_{i=1}^{N} \left|I^{\text{ref}}(z_i) - I_N^{\text{MoM}}(z_i)\right|^2}{\sum_{i=1}^{N} |I^{\text{ref}}(z_i)|^2}}, \tag{76}$$

and

$$\epsilon_{feed} = \frac{\left|I^{\text{ref}}(\text{feed}) - I_N^{\text{MoM}}(\text{feed})\right|}{\left|I^{\text{ref}}(\text{feed})\right|}. \tag{77}$$

where I_N^{MoM} is the profile of our MoM solution for given N, while $I^{\text{ref}}(\text{feed})$ and $I_N^{\text{MoM}}(\text{feed})$ represent the feed current for the two distributions considered.

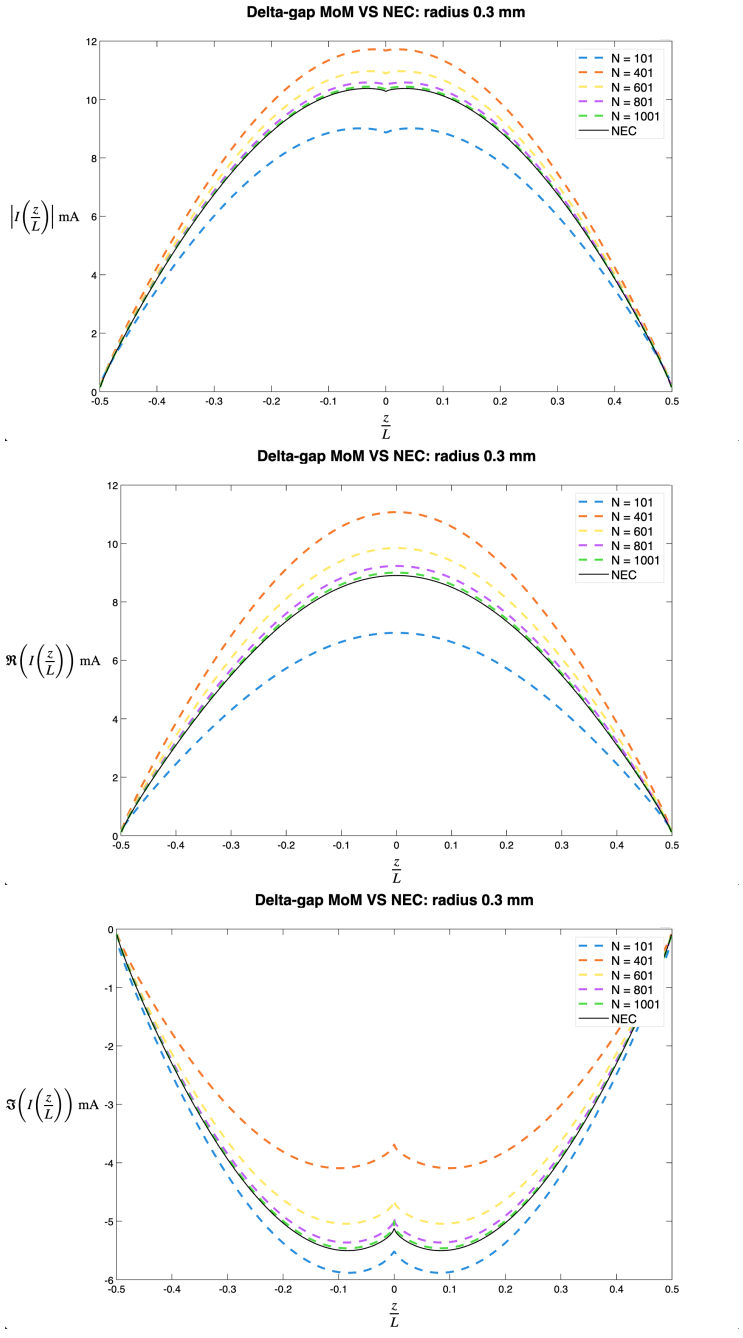


Figure 13: Comparison on NEC2 and MoM currents for different levels

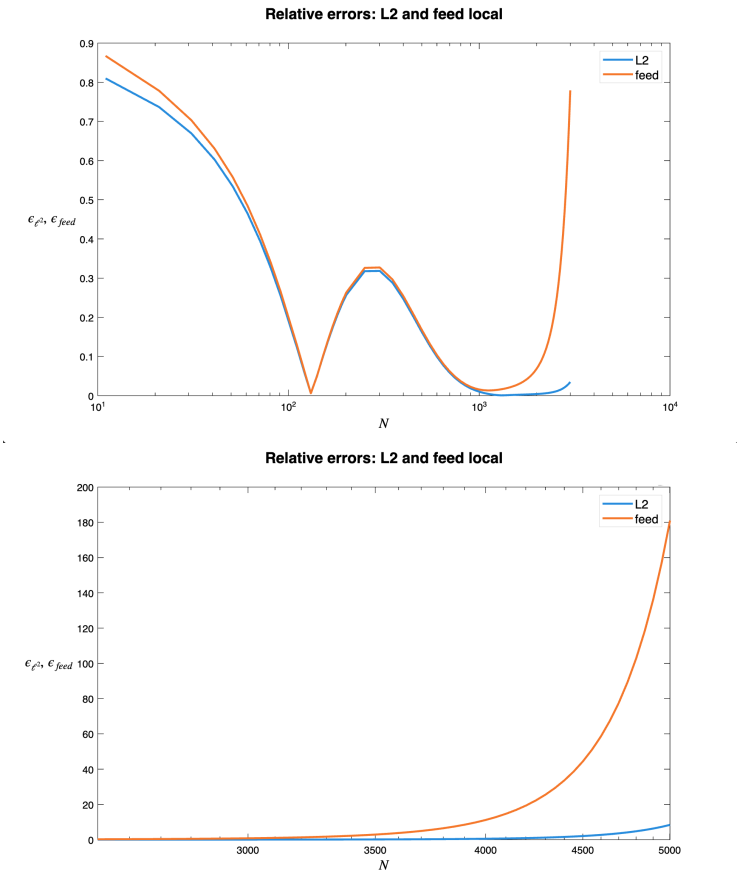


Figure 14: Relative errors

We observe that the two curves of the two relative errors are quite superposed for values on N between $N_1 \approx 100$ and $N_2 \approx 1000$ and that the feed-error ϵ_{feed} becomes exceptionally large with respect to the ℓ^2-error for higher values of N. This instance agrees with our previous analysis of the feed issues of the delta-gap source model but throws new light upon the discrete ℓ^2-norm behaviour of the solution provided by our MoM solver. The question is whether the better discrete ℓ^2-norm properties of the MoM solution could cope for the worst feed behaviour in the numerical evaluation of the dipole parameters. This question will be addressed in the subsequent sections where we will compute the input resistance and the radiative parameters.

By an investigation of numerical agreement window, we obtain for the discrete ℓ^2-error the following percentages:

$$\begin{cases} \epsilon_{\ell^2} < 1\% & \text{for} \quad 131 \leq N \leq 133 \quad \text{and} \quad 1001 \leq N \leq 2483, \\ \epsilon_{\ell^2} < 5\% & \text{for} \quad 123 \leq N \leq 141 \quad \text{and} \quad 729 \leq N \leq 3137, \end{cases} \tag{78}$$

while for the feed-error the following:

$$\begin{cases} \epsilon_{feed} < 1\% & \text{for} \quad 131 \leq N \leq 133, \\ \epsilon_{feed} < 5\% & \text{for} \quad 125 \leq N \leq 141 \quad \text{and} \quad 743 \leq N \leq 1855, \end{cases} \tag{79}$$

We notice that there is a very narrow window around $131 \leq N \leq 133$ where both ϵ_{ℓ^2} and ϵ_{feed} are below 1%. The feed-error never reaches values below 1% in other regions. Therefore, by taking into account the robustness of the discretization in N, the best candidate region of agreement for our MoM must be taken around $N \approx 1001$. For the agreement region, the lower bound is chosen as the lower bound corresponding to $\epsilon_{\ell^2} < 5\%$ in the larger region, while for its upper bound, that corresponding to $\epsilon_{feed} < 5\%$ is chosen (far away from the local minimum at $N \approx 131$). Therefore, our numerical agreement region, for the configuration under analysis, will be

$$\mathcal{N}_{5\%}(0.3 \text{ mm}) = \{N \in [737, 1855]\}. \tag{80}$$

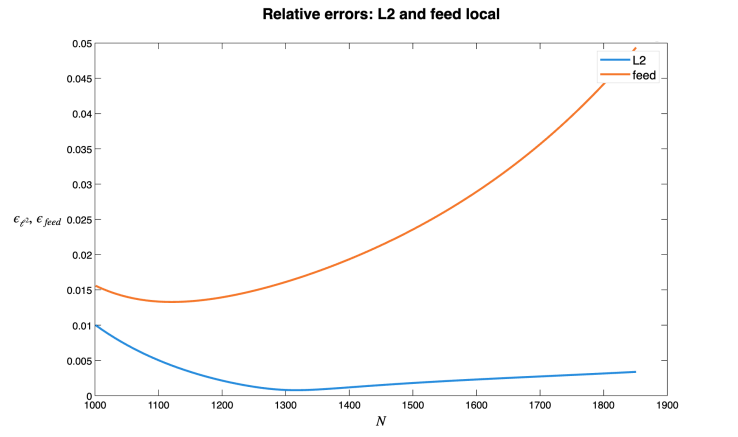


Figure 15: Agreement region $\mathcal{N}_{5\%}(0.3 \text{ mm})$

From the picture it is evident that ϵ_{feed} shows a minimum around $N \approx 1100$ while ϵ_{ℓ^2} around $N \approx 1300$, showing that the relative errors cannot be simultaneously reduced at their minimum values. We can precisely obtain the interval between the occurrence of these two minimums, which we will call $\mathcal{N}_{\min}$. For the model and the setting under investigation, this interval is given by

$$\mathcal{N}_{\min}(0.3 \text{ mm}) = \{N \in [1121, 1317]\}. \tag{81}$$

In this region, we have $\epsilon_{\ell^2} < 0.5\%$, $\epsilon_{feed} < 2\%$ and $\mathcal{N}_{\min}(0.3 \text{ mm}) \subset \mathcal{N}_{5\%}(0.3 \text{ mm})$.

We remark that Eqs.(78,79) and the ranges $\mathcal{N}_{5\%}(0.3 \text{ mm})$ and $\mathcal{N}_{\min}(0.3 \text{ mm})$, have been obtained for the reference current I^{ref}, that is, the NEC2 current for $N = 371$ segments and delta-gap excitation. However, the relative errors of the other NEC2 current profiles considered here ($N = 51, 171, 271, 471, 571, 671$) with respect to the I^{ref} are comparatively small: they do not exceed 0.7% in any case, and they decrease for higher refinements.

If we call I_0^{ref} some other NEC2 reference among the ones we have considered in this work, and we call $\epsilon_{\ell^2,0}$ and $\epsilon_{feed,0}$ the relative errors defined in Eqs.(76,77) where we change I^{ref} with I_0^{ref}, we then must have

$$\epsilon_{\ell^2,0}(N) \leq \epsilon_{\ell^2}(N) + 0.007, \qquad \epsilon_{feed,0}(N) \leq \epsilon_{feed}(N) + 0.007, \tag{82}$$

hence a relatively small change in the individuation of the numerical agreement region. The bottleneck of these bounds is the profile at $N = 51$.

3.3.2 Delta-gap validation: $a = 3$ mm

The following pictures represent the profiles of the current distribution given by NEC2. It is seen that for $N = 171$ on, the solution exhibits the pathological behaviour at the feed of the delta-gap source model. Things are much better behaved for smaller values of the discretization parameter N, and hence we can regard the region $51 \leq N \leq 111$ as the one where we could pick up the reference current I^{ref}. For subsequent analysis, we would consider I^{ref} as the NEC2 current for $N = 51$.

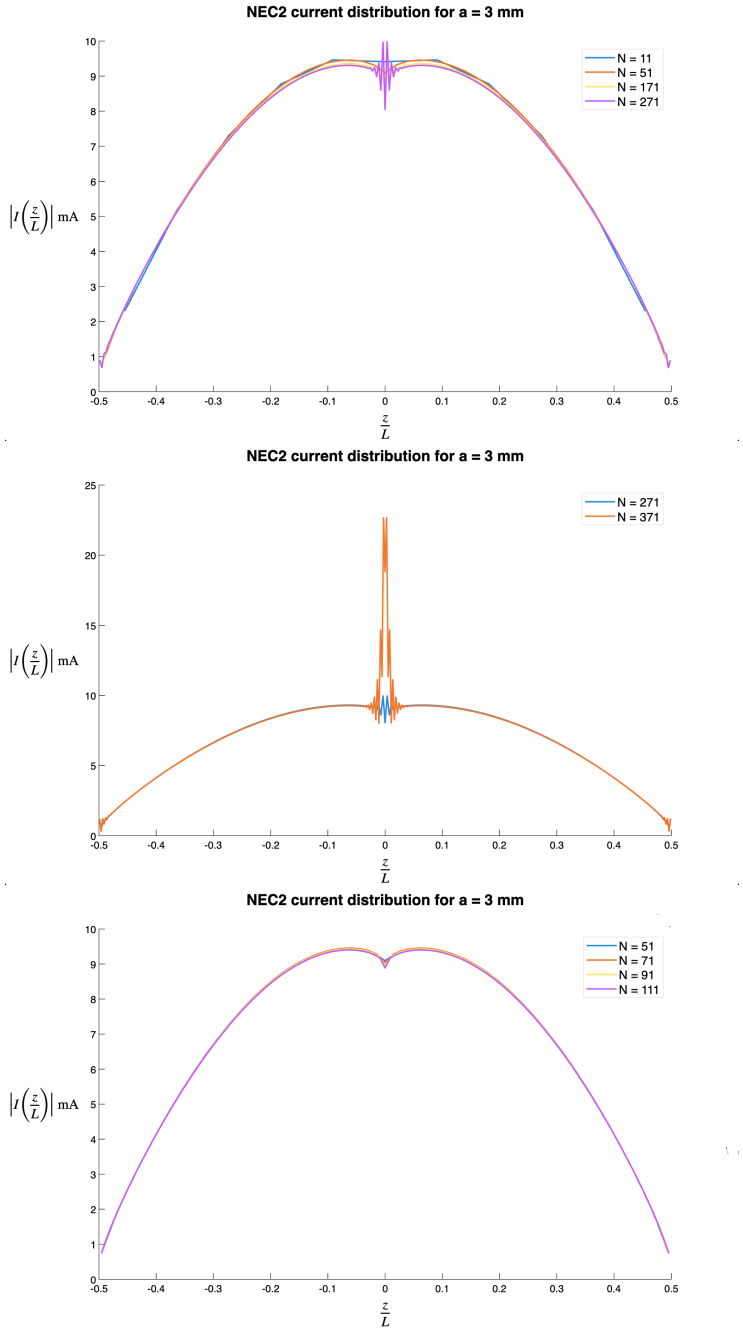


Figure 16: NEC2 current profile

In the following pictures it is shown how our MoM solution behaves with

respect to I^{ref}:

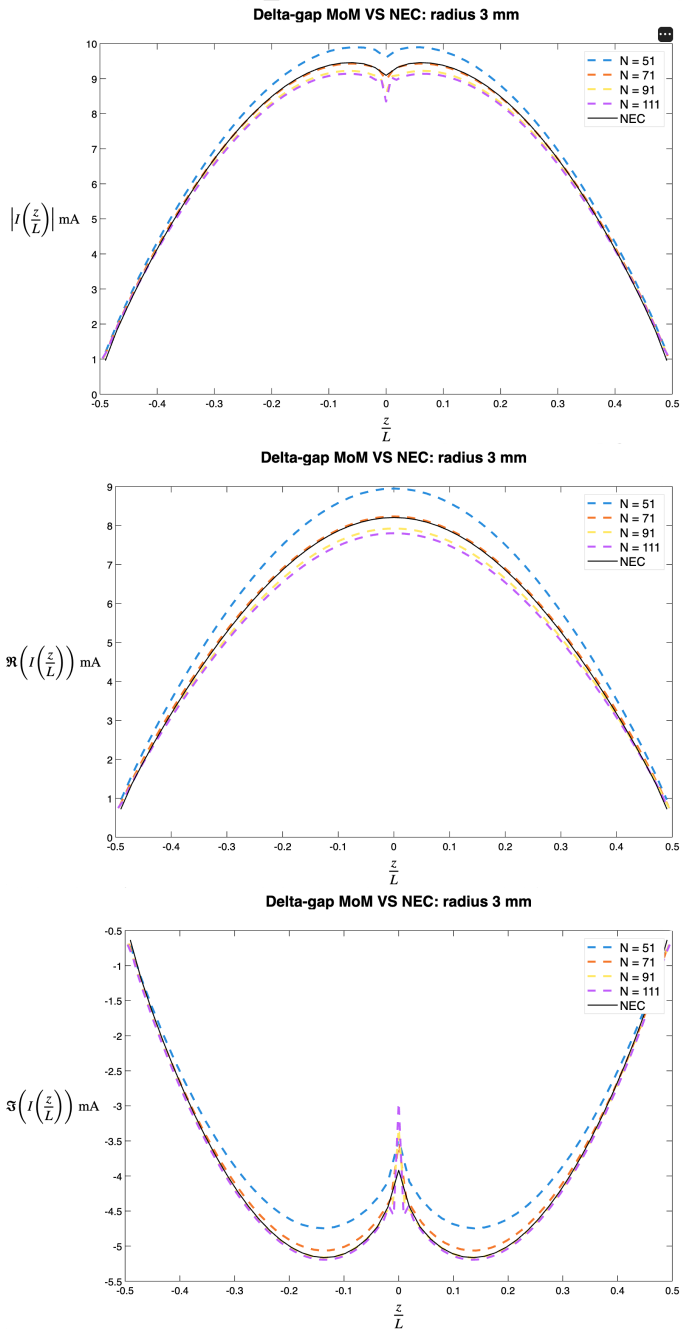


Figure 17: Comparison of NEC2 and MoM currents for different levels

We next compute the errors in Eqs.(76,77) for the configuration considered here:

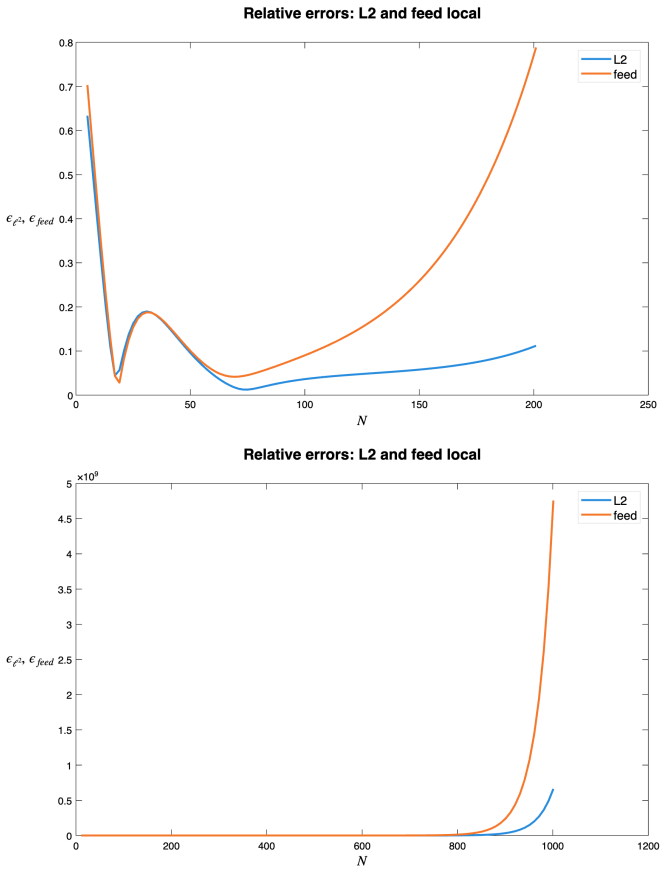


Figure 18: Relative errors

These curves are superposed between $N \approx 5$ and $N \approx 60$ and, as it happens in the case of $a = 0.3$ mm, the feed-error increases much faster than the ℓ^2-error.

By an investigation of numerical agreement window, we obtain for the discrete ℓ^2-error the following percentages:

$$\begin{cases} \epsilon_{\ell^2} < 1\% & \text{no } N, \\ \epsilon_{\ell^2} < 5\% \quad \text{for} \quad 17 \leq N \leq 19, \quad 57 \leq N \leq 129 \end{cases} \tag{83}$$

while for the feed-error the following:

$$\begin{cases} \epsilon_{feed} < 1\% & \text{no } N, \\ \epsilon_{feed} < 5\% \quad \text{for} \quad 17 \leq N \leq 21, \quad 61 \leq N \leq 87, \end{cases} \tag{84}$$

This circumstance is different from the case of $a = 0.3$ mm because the relative errors never reach a value below 1%. Using the same bounds as that used in the delta-gap case, our numerical agreement region for the configuration under analysis will be

$$\mathcal{N}_{5\%}(3 \text{ mm}) = \{N \in [61, 87]\} \tag{85}$$

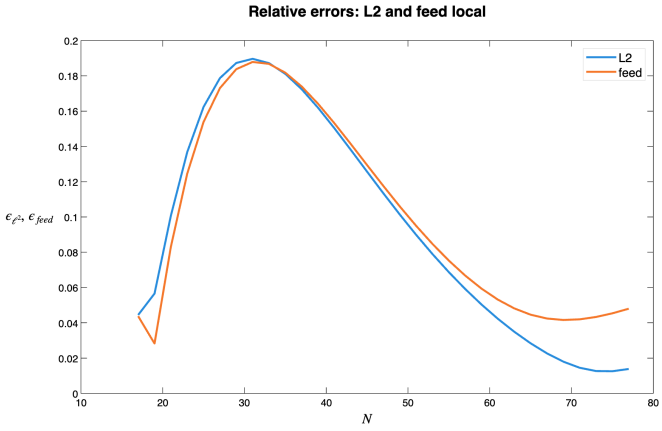


Figure 19: Agreement region $\mathcal{N}_{5\%}(3 \text{ mm})$

From this picture it is clear that a minimum of ϵ_{feed} occurs around $N \approx 67$, right before the occurrence of the minimum of ϵ_{ℓ^2}, which is likely to happen around $N \approx 75$. We can compute this window, hence the range between the two minimums is exactly given by

$$\mathcal{N}_{\min}(3 \text{ mm}) = \{N \in [69, 75]\}. \tag{86}$$

In this region we have $\epsilon_{\ell^2} < 3.5\%$, $\epsilon_{feed} < 5\%$ and $\mathcal{N}_{\min}(3 \text{ mm}) \subset \mathcal{N}_{5\%}(3 \text{ mm})$.

As for the analysis for the configuration $a = 0.3$ mm, we have obtained Eqs.(84,84) and the ranges $\mathcal{N}_{5\%}(3 \text{ mm})$ and $\mathcal{N}_{\min}(3 \text{ mm})$ for the reference current I^{ref}, that is, the NEC2 current for $N = 51$ segments and delta-gap excitation. If we set I_0^{ref} another NEC2 current among the ones we have considered here ($N = 71, 91, 111$) for the present configuration, the relative errors between I^{ref} and I_0^{ref} never exceed 3.2%. In particular, the ℓ^2-error does not exceed 0.7% while the feed-error does not exceed 3.2%. By calling $\epsilon_{\ell^2,0}$ and $\epsilon_{feed,0}$ the relative errors defined in Eqs.(76,77) where we change I^{ref} with I_0^{ref}, then we must have

$$\epsilon_{\ell^2,0}(N) \leq \epsilon_{\ell^2}(N) + 0.007, \qquad \epsilon_{feed,0}(N) \leq \epsilon_{feed}(N) + 0.035, \tag{87}$$

hence, a relatively small change for the ℓ^2-profile but a more pronounced change for the feed current. The sensitivity of the feed current seems again to be related to the discretization parameter and the radius of the wire.

3.3.3 Agreement's ranges with NEC2: delta-gap

In the previous sections, we have considered the delta-gap excitation for two configurations of wires which differ by an order of magnitude one for the other. We have derived an empirical law that describes the agreement of these configurations with respect to given reference current profiles furnished by NEC2. In particular, we have been led to consider two ranges, namely, $\mathcal{N}_{5\%}(a)$ and $\mathcal{N}_{\min}(a)$, which furnish an approximation of the numerical rage of discretization for the ℓ^2-error and the feed-error both below 5%, and the numerical range which separates the occurrence of the minimums of these errors, for $a = 0.3$ mm and $a = 3$ mm.

In this section we briefly describe the behaviour of these numerical ranges for several values of the radius a. The reference currents are always taken from NEC2 and our results must be considered as an approximation law, not as rigid mathematical proofs. This must be the case because one can arbitrarily slightly change the reference currents, which could affect the overall result.

We set the reference currents.

$$\begin{aligned} n_{5\%}(a) &= \min \mathcal{N}_{5\%}(a), \\ N_{5\%}(a) &= \max \mathcal{N}_{5\%}(a), \\ n_0(a) &= \min \mathcal{N}_{\min}(a), \\ N_0(a) &= \max \mathcal{N}_{\min}(a). \end{aligned} \tag{88}$$

We have considered thirty values of the radius in the range $a \in [0.3 \text{ mm}, 5 \text{ mm}]$. From the data obtained by this analysis, the empirical law of these bounds is given by a log-log fit as below:

$$n_{5\%}(a) \simeq 0.096833 \cdot a^{-1.1079}, \tag{89}$$

$$N_{5\%}(a) \simeq 0.036534 \cdot a^{-1.3389}, \tag{90}$$

$$n_0(a) \simeq 0.065059 \cdot a^{-1.2032}, \tag{91}$$

$$N_0(a) \simeq 0.063894 \cdot a^{-1.2155}. \tag{92}$$

These functional relations could be used whenever one needs to design a new dipole or more complex wire structures with the aid of the MoM solver for a thin wire dipole, pulse basis functions and point matching. They must be intended as guide lines without assuming their exactness. As usual, it is stressed there that these results have been obtained by considering reasonable reference currents from NEC2.

The slopes of these decaying curves are given in the figures below for the range of radius studied there, that is, $a \in [0.3 \text{ mm}, 5 \text{ mm}]$:

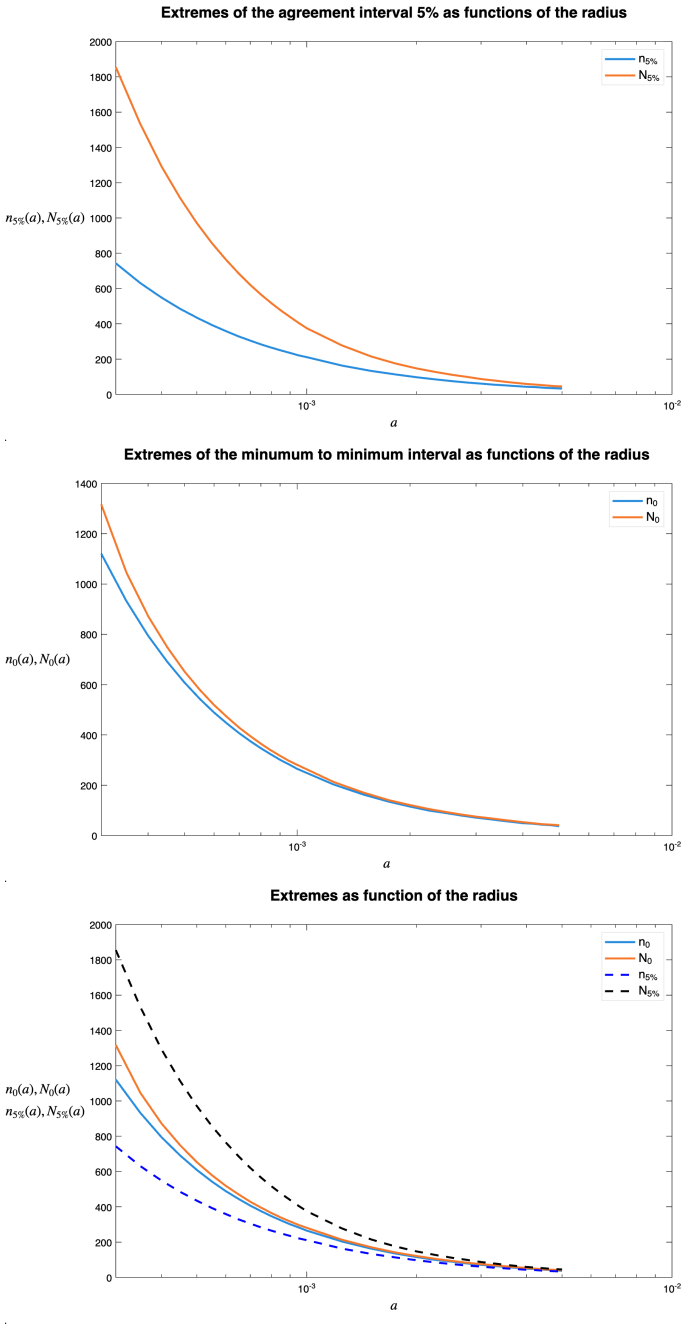


Figure 20: Extremes of the agreement intervals as functions of the radius

If we use the empirical laws derived above, we get, for $a = 0.3$ mm and $a = 3$ mm the following:

$$\begin{aligned} n_{5\%}(0.3 \text{ mm}) = 773.87, \quad & N_{5\%}(0.3 \text{ mm}) = 1903.18, \\ n_0(0.3 \text{ mm}) = 1127.29, \quad & N_0(0.3 \text{ mm}) = 1223.26, \end{aligned} \tag{93}$$

$$\begin{aligned} n_{5\%}(3 \text{ mm}) = 60.38, \quad & N_{5\%}(3 \text{ mm}) = 87.21, \\ n_0(3 \text{ mm}) = 70.60, \quad & N_0(3 \text{ mm}) = 74.48, \end{aligned} \tag{94}$$

the maximum relative error of these bounds, with respect to the results obtained in the previous sections, is almost of 7.12% (corresponding to the upper bound of the minimum-to-minimum window for $a = 0.3$ mm). The mean absolute error for these eight extremes is 2.44%.

3.4 Agreement's ranges with NEC2: magnetic frill

In the previous sections, we have shown that the solution provided by our MoM solver in the case of the magnetic frill excitation, behaves incredibly good at the feed location, even for large values of the discretization parameter N, but lacks of accuracy in the neighborhood of the endpoints. Nevertheless, the exact solution to EFIE of a thin wire requires that the current at the endpoints must vanish. For the present section, we will introduce the endpoints relative error normalized to the maximum of the reference current, which occurs at he feed point:

$$\epsilon_{endpt}(N) = \frac{|I^{\text{ref}}(\text{endpt}) - I_N^{\text{MoM}}(\text{endpt})|}{|I^{\text{ref}}(\text{feed})|}, \tag{95}$$

and consider the ℓ^2-error as in Eq.(76) along with it.

Because the delta-gap excitation provides a really good solution at the end points, we can certainly compare our MoM magnetic frill excitation with the NEC2 references also used in the previous sections, in the at least in neighborhood of the endpoints. Moreover, even though NEC2 does not provide the possibility to implement the magnetic frill excitation, we will anyway compare the MoM magnetic frill with NEC2 in terms of the local error ℓ^2. This choice is completely arbitrary and could serve as a guide line to empirically determine the ranges of agreement between our MoM magnetic frill and the delta-gap NEC2 solutions.

The next graph provides the slopes of the agreement ranges of discretization: the 5% window and the min-to-min window.

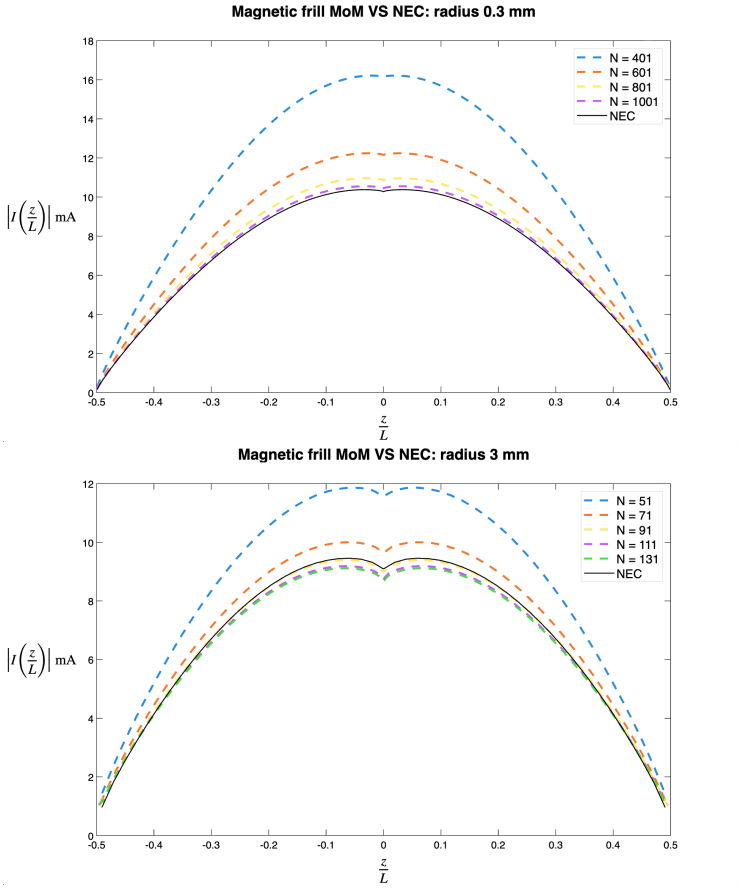


Figure 21: Comparison of NEC2 and MoM currents

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[3] Walt C. Gibson. *The Method of Moments in Electromagnetics*. CRC Press, 2022.